

Test Overview and Instructions

This assessment evaluates your ability to build, simulate, and interpret a biological circuit using SimBiology. The test is completed individually during your scheduled tutorial period.

- **Duration:** 2 hours
- **Tool:** MATLAB SimBiology
- **Format:** Individual, in-tutorial assessment
- **Submission:** Via Avenue to Learn

This test is designed to allow students with different levels of confidence and experience to demonstrate their understanding through a tiered, level-based structure.

Required Equipment

- You must bring your own laptop
- Ensure your laptop is:
 - Fully charged before the test
 - Capable of running MATLAB and SimBiology
- Power outlets are not guaranteed

Technical issues due to an unprepared laptop will not be accommodated during the test.

Permitted Resources

You may use any resources, including:

- Lecture notes
- Tutorials
- MATLAB / SimBiology documentation
- Textbooks
- Online reference material

Prohibited Activity

- Any form of communication is strictly prohibited, including:
 - Email
 - Text messages
 - Messaging apps
 - Chat platforms
 - Collaboration or discussion with other students

Academic Integrity Notice

If a student is found communicating during the test, they will receive:

- A score of zero on the assessment
- A charge of academic dishonesty in accordance with the McMaster Academic Integrity Policy

Level-Based Structure

The test consists of four levels of increasing difficulty. Each level builds directly on the previous one. You may start at any level and attempt levels in any order. However, higher levels assume all lower-level components are correctly implemented.

Maximum Grades by Level

Level Completed	Maximum Grade
Level 1	50%
Level 2	70%
Level 3	85%
Level 4	100%

How Levels Work

- All levels use one continuous SimBiology model
- Each level adds:
 - New components
 - New reactions
 - New regulatory features
- Model construction is cumulative:
 - Work done at lower levels directly carries forward to higher levels
 - Building lower-level components does not need to be repeated
- Completing a lower level may take additional time only because you must answer the questions associated with that level.
- The model-building itself does not slow down progress toward higher levels, since higher-level circuits reuse the same model and only require incremental additions.

Example strategies include:

- Completing Level 1 first to secure 50%, then attempting higher levels
- Starting directly at Level 3 or Level 4 if confident
- Completing levels sequentially

All approaches are acceptable.

Questions and Grading

Each level contains 5 questions, which may include:

- Numerical outputs
- Simulation plots
- Short interpretation questions

Grading Rules

- A level is graded only if it is completed, i.e. all questions in that level are attempted
- Partial completion of a higher level does not count
- Partial credit within a question may be awarded

If you complete multiple levels, grading will proceed as follows:

1. The highest completed level will be graded first.
2. If the score for that level is lower than the maximum possible score of the next lower completed level, the lower completed level will be graded.
3. This comparison continues until the final grade is determined.
4. The final grade will be the highest grade achieved in a completed level.

Attempting higher levels cannot reduce your final grade.

Parameters and Simulation Settings

To ensure consistency:

- All model parameters will be provided
- All simulation settings will be provided, including:
 - Simulation time
 - Initial conditions
 - Solver settings (if applicable)

You are expected to use these values exactly.

Errors and Partial Credit

Minor Errors (Partial Credit Possible)

Examples:

- Small numerical deviations
- Minor plotting or labeling issues
- Small parameter entry mistakes that do not alter circuit logic

Major Errors (Little or No Credit)

Examples:

- Missing required components or reactions
- Incorrect regulatory logic
- Plotting the wrong species
- Omitting required elements of the circuit

Submission Instructions

You must submit two files on Avenue to Learn to the SimBiology Lab Test Folder:

1. One PDF containing all required plots and written responses. The PDF should have your full name, the test version number, and the level and question numbers should be clearly indicated.
2. One SimBiology project file (.sbproj)

Students must return test paper to invigilators before leaving the test.

Naming Conventions

Name your SimBiology model: X_Detector

Both files must be named exactly as follows:

LastName_FirstName_Version#_SimBiologyTest

Example:

Doe_Jane_Version1_SimBiologyTest.pdf

Doe_Jane_Version1_SimBiologyTest.sbproj

Failure to follow the naming convention will result in a 5% penalty.

Scenario

Disease X is associated with Biomarker X and elevated temperature ($> 40^{\circ}\text{C}$) in infected areas. You have designed the following biological system to detect Disease X.

- Temp promoter is activated above 40°C , transcription rate should be modeled using the logistics model
- Toehold Switch X is activated in the presence of Biomarker X
- Enzyme B catalyzes the reaction of Protein C $\rightarrow$ Product D, assume Michaelis-Menten kinetics without inhibition
- Promoter AD is activated in the presence of AD complex. AD complex is formed by the complexation of Protein A and Product D.
- Assume all mRNA have the same loss rate: $k_{\text{mRNA_loss}}$
- Assume all protein, enzyme, product, complex, and signal species have the same loss rate: $k_{\text{pro_loss}}$
- Biomarker X concentration is assumed to be at steady state and there is no loss rate.
- Assume no cooperativity in transcription or translation. i.e Hill's Coefficient (h) = 1.
- Assume all species have an initial concentration of 0.
- All species should have units of molarity.

Below are the SBOL diagram and the table of parameters.

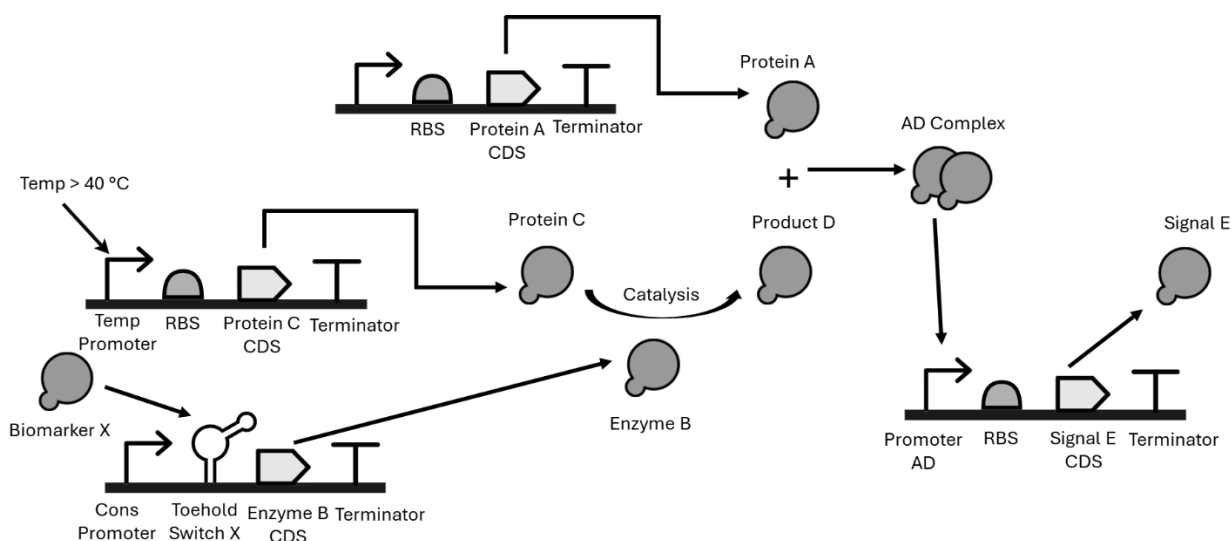


Figure 1 SBOL Diagram of the biological circuit for the detection of Disease X

Table 1 Parameter values and description for the biological circuit

Parameter	Value	Unit	Description
k_{trsl}	0.9256	1/minute	Translation rate constant for RBS
$k_{\text{mRNA_loss}}$	0.1386	1/minute	mRNA loss rate constant for all mRNA
$k_{\text{pro_loss}}$	0.0231	1/minute	Protein loss rate constant for all proteins
k_{trsc}	7.08E-09	molarity/minute	Transcription rate constant for all mRNA unless otherwise stated

K_X	6E-08	molarity	Equilibrium dissociation constant for Biomarker X and Toehold switch X
Km	1E-05	molarity	Michaelis-Menten constant for Enzyme B
kcat	6	1/minute	Catalysis rate constant for Enzyme B
K_AD	6E-08	molarity	Equilibrium dissociation constant for AD Complex and Promoter AD
k_trsl_B	7.08E-11	molarity/minute	Translation rate constant for Enzyme B
Temp_half	310	kelvin	Temperature where Temp Promoter is at half of maximum activity
Sigma	3	Unitless	Logistic model constant for Temp Promoter
k_trsc_C	7.08E-25	molarity/minute	Transcription rate constant for Temp Promoter.
k_a	1000	1/minute/molarity	Association constant for AD Complex
k_d	1E-03	1/minute	Dissociation constant for AD Complex
Cell Volume	1e-16	liter	Volume of Cell

Level 1 (Maximum Grade: 50%)

Model the portion of the circuit shown below.

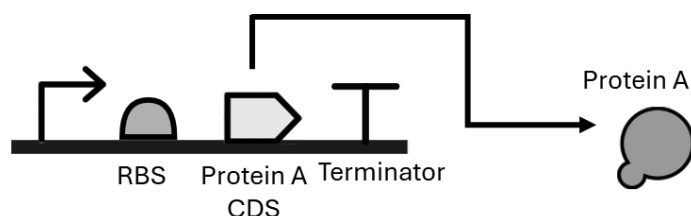


Figure 2 SBOL Diagram for Level 1

Simulate the circuit for 1000 minutes and answer the following questions. Each question is worth 20% of the maximum grade of the level.

1. What is the steady state concentration of Protein A and approximately when does it reach steady state?
2. What is the steady state concentration of the mRNA of Protein A and approximately when does it reach steady state?
3. Export a graph showing Protein A and the mRNA of Protein A as a function of time. They should be plotted as separate sets. Make sure that the graph is properly labeled and captioned.
4. What are the steady state concentrations of Protein A and mRNA of Protein, if the transcription rate constant is decreased by one order of magnitude (i.e. decrease by 10x).
5. What are the steady state concentrations of Protein A and mRNA of Protein, if the translation rate constant is increased by one order of magnitude (i.e. increased by 10x).

Level 2 (Maximum Grade: 70%)

Model the portion of the circuit shown below.

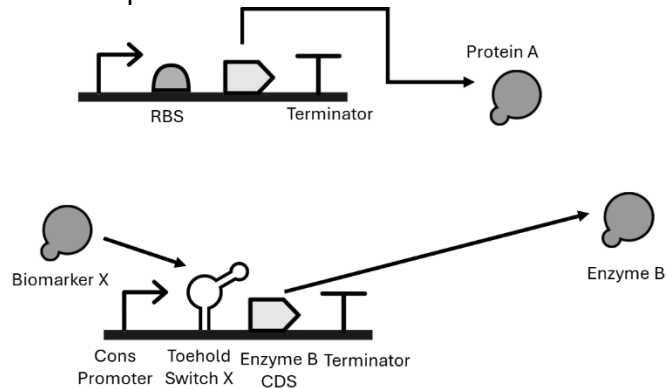


Figure 3 SBOL Diagram for Level 2

Simulate the circuit for 1000 minutes and answer the following questions. Each question is worth 20% of the maximum grade of the level.

1. What is the steady state concentration of Protein A and approximately when does it reach steady state?
2. When Biomarker X has a concentration of 5 nM, what is the steady state concentration of Enzyme B and approximately when does it reach steady state?
3. When Biomarker X has a concentration of 100 nM, what is the steady state concentration of Enzyme B and approximately when does it reach steady state?
4. Export a graph showing Protein A and Enzyme B concentrations as a function of time. They should be plotted as separate sets. The graph should show Protein A and Enzyme B concentrations when Biomarker X concentration is at 5 nM (solid line) and 100 nM (dotted line). Make sure that the graph is properly labeled and captioned.
5. What are the steady state concentrations of Enzyme B at when Biomarker X is at 5 nM and when Biomarker X is at 100 nM, if the equilibrium dissociation constant (K_x) is decreased by one order of magnitude (i.e. decrease by 10x)?

Level 3 (Maximum Grade: 85%)

Model the portion of the circuit shown below.

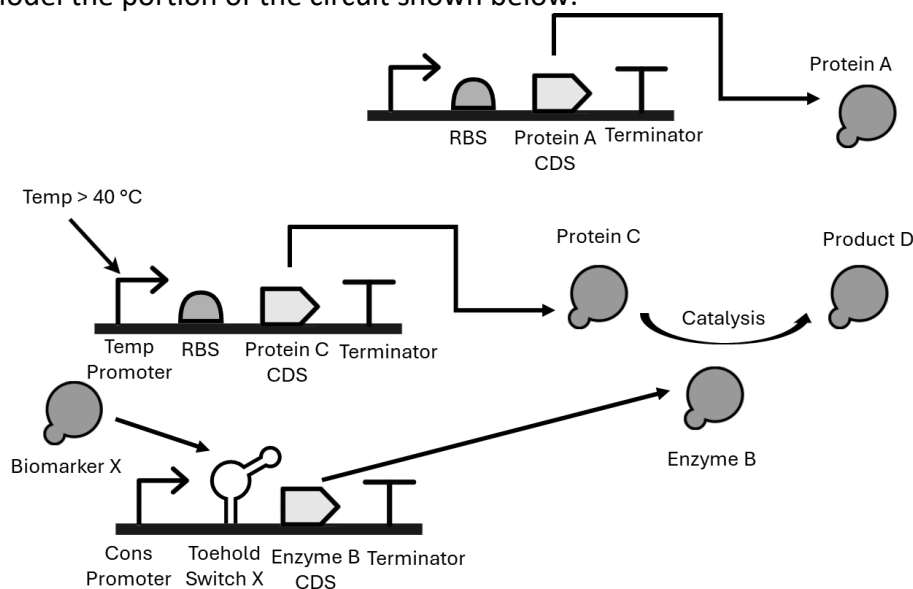


Figure 4 SBOL Diagram for Level 3

Simulate the circuit for 1000 minutes and answer the following questions. Each question is worth 20% of the maximum grade of the level. Assume conversion from °C to K is $K = ^\circ\text{C} + 273$.

1. What is the steady state concentration of Protein A and approximately when does it reach steady state?
2. When the temperature is 30 °C and Biomarker X has a concentration of 5 nM what is the steady state concentrations of Enzyme B and Product D and approximately when do they reach steady state respectively?
3. When the temperature is 40 °C and Biomarker X has a concentration of 100 nM what is the steady state concentrations of Enzyme B and Product D and approximately when do they reach steady state respectively?
4. Export a graph showing Enzyme B and Product D concentrations as a function of time. They should be plotted as separate sets. The graph should show concentrations of the two species at four conditions:

Condition	Biomarker X	Temperature
1	5 nM	30 °C
2	100 nM	30 °C
3	5 nM	40 °C
4	100 nM	40 °C

Show all four conditions on the same graph. Make sure that the graph has a legend and are properly labeled and captioned.

- What is the steady state concentration of Product D when the temperature is 40 °C and Biomarker X has an concentration of 100 nM, if the Michaelis-Menten constant is decreased by one order of magnitude (i.e. decrease by 10x)?

Level 4 (Maximum Grade: 100%)

Model the portion of the circuit shown below.

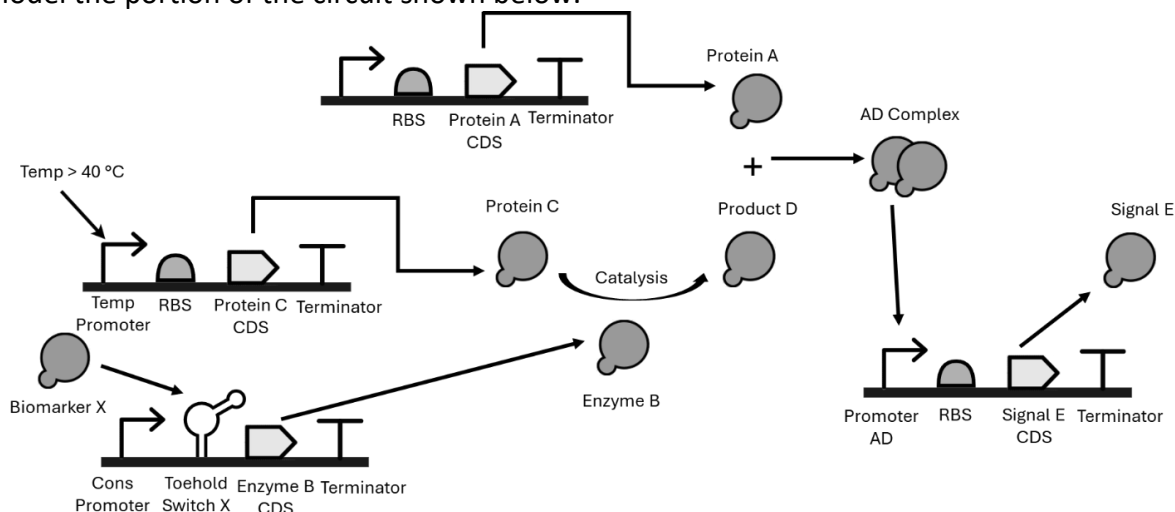


Figure 5 SBOL Diagram for Level 4

Simulate the circuit for 1000 minutes and answer the following questions. Each question is worth 20% of the maximum grade of the level. Assume conversion from °C to K is $K = ^\circ\text{C} + 273$.

- When the temperature is 30 °C and Biomarker X has a concentration of 5 nM what is the steady state concentrations of Enzyme B and Signal E, and approximately when do they reach steady state respectively?
- When the temperature is 40 °C and Biomarker X has a concentration of 100 nM what is the steady state concentrations of Enzyme B and Signal E, and approximately when do they reach steady state respectively?
- Plot and export a graph showing Enzyme B and Signal E concentrations as a function of time. They should be plotted as separate sets. The graph should show concentrations of the two species at four conditions:

Condition	Biomarker X	Temperature
1	5 nM	30 °C
2	100 nM	30 °C
3	5 nM	40 °C
4	100 nM	40 °C

IBEHS 2PB2: SimBiology Modeling Test: Version Example

Show all four conditions on the same graph. Make sure that the graph has a legend and are properly labeled and captioned.

4. Plot and export a graph showing Enzyme B and Signal E concentrations as a function of time. The initial temperature of the system is 40 °C and the Biomarker X concentration increases from an initial concentration of 5 nM to a final concentration of 100 nM at 100 minutes. The Biomarker X concentration increase is a step increase at 100 minutes after the start of the simulation. Plot Enzyme B and Signal E D as separate sets.
5. What is the steady state concentration of Signal E at a Biomarker X concentration of 100 nM and a temperature of 40 °C, if the equilibrium dissociation constant (K_x) is decreased by one order of magnitude (i.e. decrease by 10x)?

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THE END